2.4.3.2 Variable importance analysis and the origin of the landslides

To investigate the impact of each separate predictor variables on landsliding, we calibrated a univariate susceptibility model of which the AUC value is a metric for the importance of the one used variable for landslide occurrence (Figure 4). This approach is commonly used in the literature (Reichenbach et al., 2018) and was already applied successfully by Depicker et al. (2020b) for the Lake Kivu region.

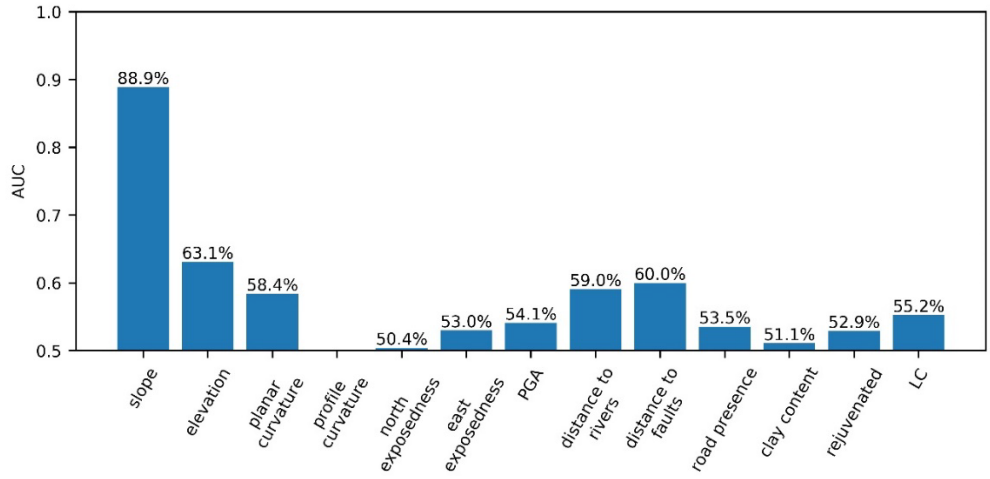


Figure 4: *Variable importance in the logistic regression model*

Figure 4 shows that slope is the main explanatory variable, followed by elevation, distance to fault, etc. The role of slope is not unexpected, as relief is the ultimate driver of landslide activity (Schmidt and Montgomery,1995), especially in hilly and mountainous landscapes such as those of Rwanda. The role of elevation could be explained by the general increase of convective rainfall with altitude. Depicker et al. (2020a, 2020b) showed that rainfall thresholds for landsliding are exceeded more often at higher elevations in the Lake Kivu region. At first sight, it might seem strange that distance to faults and PGA, proxies for seismicity, seem to play a role, despite our observation that all recent landslides were rainfall-triggered. A possible reason for the high importance of these variables could be the role of seismo-tectonic activity as a preparatory factor for landsliding, rather than a triggering factor. Seismicity weakens the hillslope material and hence reduce the minimum critical area for landslide initiation (Depicker et al., 2020a). Distance to river is certainly to be explained with presence of knickpoints and the continuous adaptation of the hillslope to river incision (Bennett et al., 2016).

The other variables play a minor role. It is however important here to keep in mind that the inventory used in the analysis was built from Google Earth images. As stressed in the inventory section, this implies that many of the very small landsides that are found along the roads are not mapped; which could be a reason why the role of roads on the occurrence of landsliding is less highlighted through the susceptibility assessment. Another reason of this limited role of roads is certainly to be found in the resolution of the topographic information used to derive the predictors. Road landslides are linked to engineered slope controls that cannot be constrained from topographic products such as the 10 m and 30 m resolution DEM (Jacobs et al., 2018; Dewitte et al., 2021).